TITLE

- **Project Name:** SHEmove
  (Women's Safety platform)
- **Team name:** Eclipse
- **Institution:** Uttarakhand Technical University
- **Team Members:** Devesh Rawat(Leader)
  Pranjal Arya
  Deepika
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IDEA OVERVIEW

Problem

Women's safety is fragmented, reactive, and disconnected.

One crime against women every 41 sec is registered as of 2025 data, yet existing systems still depend on victims actively seeking help.

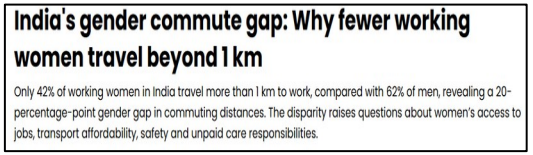


94% women feel unsafe travelling alone in India, survey finds

Himanshi Dhawan / TNN / Feb 26, 2013, 05:08 IST

The Delhi gang-rape incident has not just bruised the image of the national Capital, but the country as well. A travel survey has found that 94% women feel unsafe traveling alone, while 84% voted for Delhi as the most unsafe metro.

NEW DELHI: The Delhi gang-rape incident has not



India's gender commute gap: Why fewer working women travel beyond 1 km

Only 42% of working women in India travel more than 1 km to work, compared with 62% of men, revealing a 20-percentage-point gender gap in commuting distances. The disparity raises questions about women's access to jobs, transport affordability, safety and unpaid care responsibilities.

Unique Selling Point

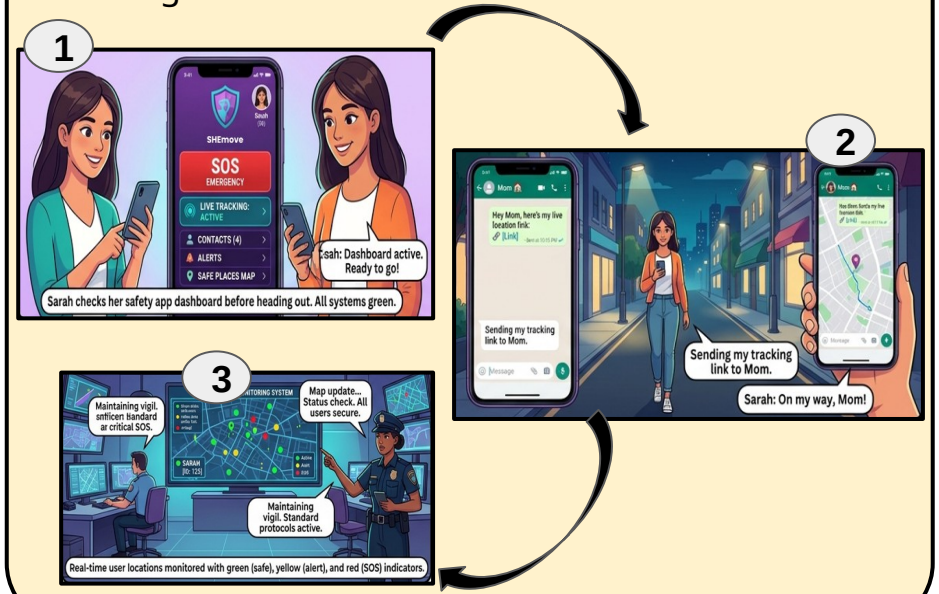
Dual-PIN Convert Distress Authentication

Regular PIN:
Normal authentication and app access.

Duress PIN:
Covert emergency activation and location sharing.

Our Solution

An integrated proactive women safety platform that connects women, emergency responders, and authorities through real-time SOS alerts, live location tracking, and intelligent incident management.



TECHNICAL APPROACH

System Architecture and Workflow

1 Woman's App

Register
Sign up and add trusted contacts

Plan Journey
Destination
Estimated travel time
Check-in frequency

Dual-PIN Security
- **Regular PIN** — Edit or extend journey
- **Duress PIN** — Looks normal, sends silent alert

2 Journey Monitoring

Live Location
Tracked securely during the journey

Scheduled Check-ins
Configurable timer asks "Are you safe?"

Anomaly Detection
Missed check-in
Network disconnection

Escalation Engine
Decides Green, Yellow or Red status

3 Safety Status

GREEN
Normal Journey
Movement shown only, no identity revealed

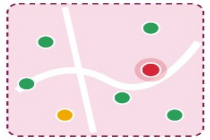
YELLOW
Potential Anomaly
Missed check-in or network drop

RED
SOS / Duress Alert
Emergency escalated to police for response

4 Dashboards

Trusted Contact Dashboard
WhatsApp journey link
Live progress and status
No app install needed

Police Dashboard
Map of one station's jurisdiction
- Anonymous marker
- Last known location and journey stats
- Full incident details to respond



Duress PIN goes straight to RED, silently

Privacy and Role-Based Access

Woman	Full control of her journey
Contact	Journey progress and status only
Police	Anonymous until RED alert

Future Scope

- **Adaptive check-ins** — Interval adjusts using journey patterns
- **Offline relay** — Bluetooth hubs at police stations

Real-World Impact

- Proactive safety, not just an SOS button
- Privacy-first by design
- Faster, targeted police response
- Scales one jurisdiction at a time

Tech Stack

Python · Flask · Next.js · Vercel · PostGIS · React · MongoDB · Google Maps · Firebase · Tailwind CSS · Flutter · Kotlin · JavaScript

FEATURES OF PROJECT

Silent SOS Trigger:
Discreetly activate emergency protocols without alerting the aggressor.

Live Location Tracking:
Share real-time GPS coordinates with trusted emergency contacts.

Dual-PIN Authentication:
Separate regular and duress PINs for normal access and covert SOS activation.



Smart Threat Detection:
Detect potential distress through contextual audio and sensor signals.

Emergency Escalation:
Automatically notify predefined contacts during distress situations.

Evidence Capture:
Securely capture and preserve relevant audio evidence during emergencies.

Safe Route Navigation:
Identify safer routes and highlight nearby emergency assistance points.

SHEmove

BUSINESS MODEL

Team Eclipse

Key Partners

- *State Police & Emergency Response Agencies*
- *1,168 Universities*
- *45,473 Colleges*
- *2.19 Million Active Companies*
- *₹7,712.85 Cr Nirbhaya Fund allocation*

Key Activates

- *2-PIN Authentication System*
- *99.5% Target Platform Uptime*

Key Resources

- *1 Unified Safety Platform*
- *3 Revenue Channels*

Value Preposi

- *4,41,534 Registered Crimes Against Women (2024)*
- *1,210 Cases Registered Daily*
- *2 Authentication Modes*
- *1 Covert Emergency Trigger*

Customer Relationship

- *1:1 Institutional Onboarding*

Channels

- *2.19 Million Active Companies Potential*
- *2 Major App Stores*

Customer Segment

- *70.9 Cr Female Population*
- *2.07 Cr Female Higher-Education Enrolments*
- *2.19 Million Active Companies*
- *45,473 Colleges*

Cost Structure

- *₹30/User/Month — Proposed SaaS Cost Basis*
- *Cloud & Infrastructure Costs*
- *Deployment & Maintenance Costs*

Revenue Streams

- *₹49–99/Month — B2C Subscription*
- *₹99–149/Month — Family Plan*
- *Annual Government Licensing & AMC*

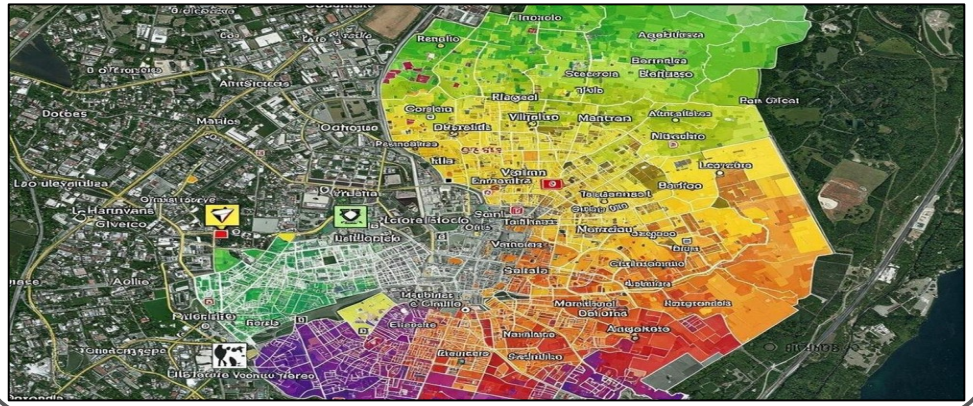
Community-Driven Safety Heatmap

1. **User Feedback:** Women anonymously report areas where they feel unsafe.

2. **Data Aggregation:** Reports are grouped by geographical location.

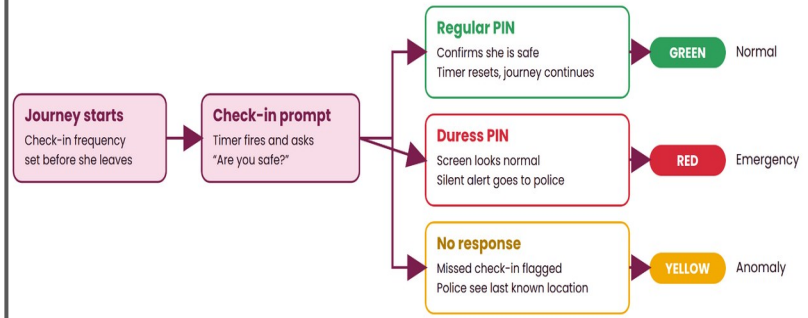
3. **Risk Mapping:** Areas receive color-coded safety indicators based on feedback volume, recency and consistency.

4. **Safer Navigation:** Users can view safety zones and choose alternative routes.



Smart Check-in

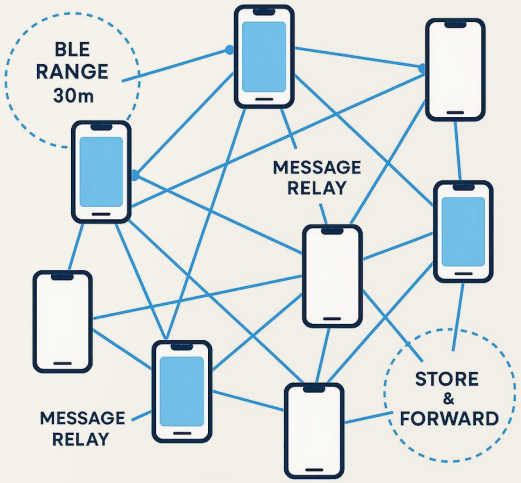
1. Journey Initiation: User enters destination and starts journey.
2. AI-Based ETA: Predicts arrival time using traffic, distance and route conditions.
3. Smart Check-in: Schedules adaptive safety check-ins.
4. Deviation Detection: Flags unexpected delays or route deviations.
5. Safety Escalation: Prompts confirmation and triggers emergency escalation if needed.



FUTURE INNOVATION

Bluetooth Mesh Hopping:

Enables nearby participating devices to relay emergency signals through Bluetooth when conventional internet connectivity is unavailable, extending distress communication beyond a single device's range.



Voice-Code Activation

Uses personalized, predefined keywords (e.g. "Charlie") to trigger a covert SOS, allowing hands-free emergency activation without touching the phone.



Features	112 India	My Safetipin	SHEmove
Emergency SOS	✓	–	✓
Live location tracking	✓	✓	✓
Proactive risk prevention	–	✓	✓
Community-based unsafe zone mapping	–	✓	✓*
Duress / Dual PIN	–	–	✓
Custom voice-code SOS	–	–	Planned
Bluetooth SOS relay	–	–	Planned
User-reported danger zones	–	✓	Planned
Multi-layered safety approach	Reactive	Preventive	Both



"To every woman who has ever shared her location just to feel safe; we hope one day you won't have to."

~Team Eclipse